

Complex solutions to quadratic equations

Basic skills

Q1. a) $\sqrt{-4} = \sqrt{-1} \times \sqrt{4} = i \times 2 = 2i$

b) $\sqrt{-6} = \sqrt{-1} \times \sqrt{6} = i\sqrt{6} = \sqrt{6}i$

c) $\sqrt{-8} = \sqrt{-1} \times \sqrt{8} = i \times 2\sqrt{2} = 2\sqrt{2}i$

d) $\sqrt{-44} = \sqrt{-1} \times \sqrt{44} = i \times 2\sqrt{11} = 2\sqrt{11}i$

Q2. a) $x^2 + 4x + 5$

$a=1 \quad b=4 \quad c=5 \quad d=b^2-4ac$

$d \begin{cases} > 0 & 2 \text{ real} \\ = 0 & 1 \text{ real} \\ < 0 & 2 \text{ complex} \end{cases}$

$d = 4^2 - 4 \times 1 \times 5 = 16 - 20 = -4$

as $d < 0$ 2 complex roots

b) $x^2 - 4x - 5$

$d = (-4)^2 - 4 \times 1 \times -5 = 16 + 20 = 36$

as $d > 0$ 2 real roots

c) $2x^2 - 3x + 4$

$$d = (-3)^2 - 4 \times 2 \times 4 = 9 - 32 = -21$$

$d < 0$ so 2 complex roots

d) $-2x^2 + 10x - 7$

$$d = 10^2 - 4 \times -2 \times -7 = 100 - 56$$

$d = 44 > 0$ so 2 real roots

Q 3.

$$x^2 - 8x - 2 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-8) \pm \sqrt{8^2 - 4 \times 1 \times -2}}{2 \times 1}$$

$$= \frac{8 \pm \sqrt{72}}{2} = \frac{8 \pm 6\sqrt{2}}{2}$$

$$= 4 \pm 3\sqrt{2}$$

Competency skills

Q 4.

$$x^2 + 9 = 0$$

$$x^2 = -9 \quad \text{so} \quad x = \pm \sqrt{-9} = \pm 3i$$

(could use quadratic formula)

Q5.

$$x^2 + 2x + 5 = 0$$

$$a=1 \quad b=2 \quad c=5$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4 \times 1 \times 5}}{2} = \frac{-2 \pm \sqrt{-16}}{2}$$

$$= \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

Q6.

$$4x^2 - 3x + 5 = 0 \text{ for roots}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 4 \times 5}}{2 \times 4}$$

$$= \frac{3 \pm \sqrt{-71}}{8} = \frac{3 \pm \sqrt{71}i}{8}$$

Q7.

$$3x^2 = x - 2 \text{ means}$$

$$3x^2 - x + 2 = 0$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 3 \times 2}}{2 \times 3}$$

$$= \frac{1 \pm \sqrt{-23}}{6} = \frac{1 \pm \sqrt{23}i}{6}$$

Q8. a) $i^2 = (\sqrt{-1})^2 = -1$

$\times i \downarrow$ b) $i^3 = i^2 \times i = -1 \times i = -i$

$\times i \downarrow$ c) $i^4 = i^3 \times i = -i \times i = -i^2 = -(-1) = 1$

$\times i \downarrow$ d) $i^5 = i^4 \times i = 1 \times i = i$

Notice it loops $\sqrt{-1}, -1, -\sqrt{-1}, 1, \dots$

Advanced skills

Q9. a) $\sqrt{-6} \times \sqrt{5} = \sqrt{-30} = \sqrt{30}i$

b) $\sqrt{-20} \times \sqrt{2} = \sqrt{-40} = 2\sqrt{10}i$

c) $\sqrt{-8} + \sqrt{-2} = 2\sqrt{2}i + \sqrt{2}i = 3\sqrt{2}i$

Q10. $2z^2 + 6z - 10$

$$z = \frac{-6 \pm \sqrt{6^2 - 4 \times 2 \times -10}}{2 \times 2}$$

$$= \frac{-6 \pm 2\sqrt{29}}{2 \times 2} = \frac{-3 \pm \sqrt{29}}{2}$$

$$\text{So } z_1 = \frac{-3 + \sqrt{29}}{2} \quad z_2 = \frac{-3 - \sqrt{29}}{2}$$

$$\operatorname{Re}(z_1) = \frac{-3 + \sqrt{29}}{2} \quad \operatorname{Re}(z_2) = \frac{-3 - \sqrt{29}}{2}$$

$$\operatorname{Im}(z_1) = 0 \quad \operatorname{Im}(z_2) = 0$$

Q11.

$$2z^2 + 6z + 10$$

$$z = \frac{-6 \pm \sqrt{6^2 - 4 \times 2 \times 10}}{2 \times 2} = \frac{-6 \pm \sqrt{-44}}{4}$$

$$= \frac{-6 \pm 2\sqrt{11}i}{4} = \frac{-3 \pm \sqrt{11}i}{2}$$

$$z_1 = -\frac{3}{2} + \frac{\sqrt{11}i}{2} \quad z_2 = -\frac{3}{2} - \frac{\sqrt{11}i}{2}$$

$$\operatorname{Re}(z_1) = -\frac{3}{2}$$

$$\operatorname{Re}(z_2) = -\frac{3}{2}$$

$$\operatorname{Im}(z_1) = \frac{\sqrt{11}}{2}$$

$$\operatorname{Im}(z_2) = -\frac{\sqrt{11}}{2}$$

Notice no i

Q12.

$$x^2 + 2x - k$$

$$x = \frac{-2 \pm \sqrt{4+4k}}{2} = \frac{-2 \pm 2\sqrt{1+k}}{2}$$

$$= -1 \pm \sqrt{1+k}$$

if $1+k < 0$ complex solutions

so when $k < -1$ x is

complex

$$\left(x = -1 \pm \sqrt{-k-1} i \right)$$

is solution

Q13.

$$kx^2 + 2x = k$$

$$kx^2 + 2x - k = 0$$

$$x = \frac{-2 \pm \sqrt{4+4k^2}}{2} = \frac{-2 \pm 2\sqrt{1+k^2}}{2}$$

$$= -1 \pm \sqrt{1+k^2}$$

as $1+k^2 > 0$ for any k , there is never a complex solution.